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🧠 SMART AIR FRESHENER v2.1 - LIVE IOT DASHBOARD

[Cycle 40]

🕒 Timestamp: 2026-07-27 10:16:39

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📡 [SENSOR METRICS (Raw & Real-time)]

🌡️ Temperature : 17.61 °C (Accuracy: ±0.5°C)

💧 Humidity : 44.55 % RH (Accuracy: ±5.0% RH)

☠️ VOC (Odor) : 5.22 ppm (Sensitivity: ±0.1 ppm)

🌸 Scent Conc. : 6.00 µg/m³ (Target: 2.0 µg/m³)

🧴 Fluid Level : 109.0 ml (Capacity: 120 ml)

🔄 Total Sprays : 32

🏠 Occupancy : 🟡 Empty

📊 Uptime : 100.0 %

🏷️ Coverage : 500.0 sq. ft.

📊 [ENVIRONMENTAL ANALYSIS]

🏷️ Classification : WORST (Critical Intervention Required)

📈 Health Score : 20.0 / 100

📋 Breakdown :

- Temp: Acceptable
- Humidity: Ideal
- VOC: Critical (Unhealthy)
- Fluid: Sufficient

- Scent: Severely Imbalanced
- Room: Unoccupied (Eco-mode)



⚡ [VIRTUAL PHYSICAL OUTPUTS (GPIO / PWM)]

(Replace print() with GPIO.output() or serial.write() for real hardware)

LED (GPIO17) : RED (Solid - CRITICAL)
Spray Relay (GPIO18) : HIGH (ON)
Spray PWM Speed : 91 / 255 (35% Duty)
Buzzer (GPIO19) : HIGH (BEEP BEEP!)
Diffuser Fan (PWM1) : HIGH (80%) -

Ventilating

📄 [HUMAN-READABLE DEVICE LOGIC]

◆ NORMAL SPRAY: Standard cycle (0.3ml) | 💡
ECO-MODE: Reducing frequency (40% fluid saved)
💜 Spraying at 0.18 ml/cycle

📈 [SYSTEM PERFORMANCE (Infused Metrics)]

✅ Odor Neutralization Efficiency : 99.5 %
💰 Fluid Saved (vs. Standard) : 40.0 % (Smart Scheduling)
🔋 Power Consumption (Standby) : 0.5 W

📖 [CONDITION HISTORY (Preferred ↔ Worst)]

PREFERRED (Excellent) : 6 times
GOOD (Comfortable) : 6 times
MODERATE (Needs Attention) : 18 times
POOR (Unpleasant) : 8 times





Power Consumption (Standby) : 0.5 W



[CONDITION HISTORY (Preferred ↔ Worst)]

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WORST (Critical Intervention Required): 2 times

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❏ SIMULATION COMPLETE - FINAL CONDITION DISTRIBUTION

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MODERATE (Needs Attention) : 18 cycles
(45.0%)

POOR (Unpleasant) : 8 cycles
(20.0%)

PREFERRED (Excellent) : 6 cycles
(15.0%)

GOOD (Comfortable) : 6 cycles
(15.0%)

WORST (Critical Intervention Required): 2 cycles (5.0%)

✅ Device successfully analyzed the spectrum from Preferred to Worst conditions.